

G.C.E. (ADVANCED LEVEL) COMBINED MATHEMATICS

Resource Paper: Principle of Mathematical Induction

Comprehensive Problem Set • 40 Structured Questions covering Series, Divisibility, Inequalities, and Recurrence Relations

Instructions to Students:

- Answer all questions using the Principle of Mathematical Induction (PMI).
- For summation and series questions, clearly verify for $n = 1$, assume true for $n = k$, and prove for $n = k + 1$.
- For divisibility and inequality problems, state the inductive hypothesis explicitly.

Part A: Summation of Series & Algebraic Identities (Questions 1 – 18)

1. Prove by induction that for all positive integers n ,

$$1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$$

2. Prove by induction that for all positive integers n ,

$$1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

3. Prove by induction that for all positive integers n ,

$$1^3 + 2^3 + 3^3 + \dots + n^3 = \left[\frac{n(n+1)}{2} \right]^2$$

4. Show by mathematical induction that

$$1 \cdot 2 + 2 \cdot 3 + 3 \cdot 4 + \dots + n(n+1) = \frac{n(n+1)(n+2)}{3}$$

for all $n \in \mathbb{N}^+$ (or positive integers n).

5. Prove that for all positive integers n ,

$$1 \cdot 2 \cdot 3 + 2 \cdot 3 \cdot 4 + 3 \cdot 4 \cdot 5 + \dots + n(n+1)(n+2) = \frac{n(n+1)(n+2)(n+3)}{4}$$

6. Prove by induction:

$$1 + 3 + 5 + \dots + (2n-1) = n^2$$

7. Prove by induction:

$$2 + 4 + 6 + \dots + 2n = n(n+1)$$

8. Establish by induction that for all positive integers n ,

$$1 \cdot 3 + 2 \cdot 4 + 3 \cdot 5 + \dots + n(n+2) = \frac{n(n+1)(2n+7)}{6}$$

9. Prove by mathematical induction that

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \dots + \frac{1}{n(n+1)} = \frac{n}{n+1}$$

10. Prove by mathematical induction that

$$\frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \frac{1}{5 \cdot 7} + \dots + \frac{1}{(2n-1)(2n+1)} = \frac{n}{2n+1}$$

11. Prove by induction that

$$\frac{1}{1 \cdot 2 \cdot 3} + \frac{1}{2 \cdot 3 \cdot 4} + \dots + \frac{1}{n(n+1)(n+2)} = \frac{n(n+3)}{4(n+1)(n+2)}$$

12. Prove by induction:

$$1 + 2 + 4 + \dots + 2^{n-1} = 2^n - 1$$

13. Prove by induction that

$$1 + 3 + 9 + \dots + 3^{n-1} = \frac{3^n - 1}{2}$$

14. Show that for all positive integers n ,

$$a + ar + ar^2 + \dots + ar^{n-1} = \frac{a(r^n - 1)}{r - 1} \quad (r \neq 1)$$

15. Prove by induction that

$$1 \cdot 1! + 2 \cdot 2! + 3 \cdot 3! + \dots + n \cdot n! = (n+1)! - 1$$

16. Prove by mathematical induction that

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots + \frac{1}{2^n} = 1 - \frac{1}{2^n}$$

17. Prove by induction that for all positive integers n ,

$$1^2 - 2^2 + 3^2 - 4^2 + \dots + (-1)^{n-1} n^2 = \frac{(-1)^{n-1} n(n+1)}{2}$$

18. Prove by induction that

$$\sum_{r=1}^n r(r+1)(r+2)(r+3) = \frac{n(n+1)(n+2)(n+3)(n+4)}{5}$$

Part B: Divisibility Problems (Questions 19 – 28)

19. Prove by mathematical induction that $3^{2n} - 1$ is divisible by 8 for all integers $n \geq 1$.
20. Prove that $2^{3n} - 1$ is divisible by 7 for all positive integers n .
21. Prove by induction that $n^3 + 2n$ is divisible by 3 for all positive integers n .
22. Prove by induction that $n^3 - n$ is divisible by 6 for all positive integers n .
23. Show that $n^5 - n$ is divisible by 5 for all positive integers n using mathematical induction.
24. Prove by induction that $7^n - 3^n$ is divisible by 4 for all positive integers n .
25. Prove by induction that $5^{2n} - 24n - 1$ is divisible by 576 (or 24^2) for all integers $n \geq 1$.
26. Prove that $3^{n+2} + 4^{2n+1}$ is divisible by 13 for all positive integers n .
27. Prove by mathematical induction that $n^3 + 3n^2 + 5n$ is divisible by 3 for all positive integers n .
28. Prove that $2^{n+2} \cdot 3^n + 5$ is divisible by 7 for all integers $n \geq 1$.

Part C: Inequalities & Advanced Proofs (Questions 29 – 37)

29. Prove by induction that $2^n > n$ for all positive integers n .
30. Prove by induction that $2^n > 2n + 1$ for all integers $n \geq 3$.
31. Prove Bernoulli's inequality: $(1 + x)^n \geq 1 + nx$ for all integers $n \geq 1$, where $x > -1$.
32. Prove by induction that $n! > 2^{n-1}$ for all integers $n \geq 3$.
33. Prove by induction that $n! > 2^n$ for all integers $n \geq 4$.
34. Using mathematical induction, prove that

$$\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \dots + \frac{1}{\sqrt{n}} > \sqrt{n}$$

for all integers $n \geq 2$.

35. Prove by induction that for all integers $n \geq 2$,

$$\frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{2n} > \frac{13}{24}$$

36. Prove by induction that $3^n < n!$ for all integers $n \geq 7$.
37. Prove that $(n!)^2 > n^n$ for all integers $n \geq 2$. (Hint: check for $n = 2$ and use induction).

Part D: Recurrence Relations & Miscellaneous (Questions 38 – 40)

38. A sequence $\{u_n\}$ is defined by $u_1 = 3$ and $u_{n+1} = 2u_n + 1$ for $n \geq 1$. Prove by mathematical induction that $u_n = 2^{n+1} - 1$ for all integers $n \geq 1$.
39. A sequence $\{a_n\}$ is defined by $a_1 = 1$ and $a_{n+1} = \sqrt{2 + a_n}$ for $n \geq 1$. Prove by mathematical induction that $a_n < 2$ for all positive integers n .
40. Let $P(n)$ be the statement that $\sum_{r=1}^n \frac{1}{(3r-2)(3r+1)} = \frac{n}{3n+1}$. Verify $P(1)$ and $P(2)$, and prove by mathematical induction that $P(n)$ is true for all positive integers n .